

CHAPTER 3: DESIGN TRADE-OFFS

This chapter presents the evaluation of the design trade-offs based on the identified constraints. It discusses economic, safety, risk, environmental, and sustainability considerations for the three design alternatives. The chapter also includes a summary of normalized values and designers' rankings to support comparison. Lastly, it explains how these trade-offs influence the final design selection and presents a sensitivity analysis to validate the results.

Design trade-off refers to a situation where improving one design characteristic may result in the reduction of another. The researchers analyzed the tradeoff of the study using the constraints in the previous chapters of this study.

3.1 Summary of Constraints

Table of Constraints:

DESIGN 1	DESIGN 2	DESIGN 3
● Accessibility and WCAG Compliance ● Feedback and Notification System Enhancement ● Multi-Step Workflow Guidance ● Cross-Device Testing Documentation ● Brand Kit Documentation	● Technical Complexity ● Hardware Limitations ● Network Dependency ● Cost and Licensing Scalability ● Data Privacy and Compliance: ● System Reliability and Maintenance ● User Experience ● Security Limitations	● Hardware and Storage Constraints ● Cost Constraints ● Software and Compatibility Constraints ● Security and Compliance Constraints ● Operational Constraints

3.2 Trade Offs

The researchers conducted trade-off analysis to systematically compare the three proposed designs by evaluating their advantages and limitations under the given project constraints. This analysis is essential in determining the most practical and efficient solution that best aligns with the system goals and available resources.

3.2.1 Tradeoff 1: Economic

Design 1: Cloud-based with RPI NAS implementation Annual Cost

PROJECT INPUTS	Design 1	Cost
Hardware Implementation	Raspberry Pi NAS based	PHP 5,000.00
Domain name	site TLD	PHP 300.00
VPN service	Tailscale VPS(free tier)	PHP 0.00
Cloud Hosting Service	Hostinger VPS	PHP 12,228.00
TOTAL		PHP 17,528.00

The total estimated cost for implementing Design 1 of the INTRAK system is PHP 17,528.00. This includes the hardware implementation of the Raspberry Pi NAS-based setup at PHP 5,000.00, which provides the core infrastructure for storing internship records and integrating with cloud services. The domain name registration costs PHP 300.00, ensuring a professional and accessible web address for students, instructors, and administrators. The VPN service uses Tailscale's free tier, contributing no additional cost, while the cloud hosting service through Hostinger VPS accounts for PHP 12,228.00, supporting reliable and scalable access to the system.

Most of the cost is attributed to cloud hosting, followed by hardware, while domain registration and the VPN service are minimal contributors. This cost distribution

demonstrates that Design 1 provides a cost-effective balance between local hardware investment and cloud-based operations

Design 2: Cloud-Based Angular–Spring Boot Web Platform for INTRAK
Annual Cost

PROJECT INPUTS	Design 2	Cost
Hardware Implementation	No hardware implementation	N/A
Domain name	.com	PHP 1,200.00
VPN service	expressVPN	PHP 5,760.00
Cloud Hosting Service	Hostgator	PHP 10,080.00
TOTAL		PHP 18,040.00

The total estimated cost for Design 2 of the INTRAK system is PHP 18,040.00. This design does not require any physical hardware, as all components are hosted in the cloud. The domain name registration for a .com site costs PHP 1,200.00, providing a professional and accessible web address for users. The VPN service uses ExpressVPN at PHP 5,760.00, ensuring secure remote access for administrators and staff. The cloud hosting service is provided by Hostgator at PHP 10,080.00, which supports the web application, database, and storage entirely in the cloud.

The distribution of costs shows that the majority of expenses come from cloud-based services, particularly hosting and VPN subscriptions. The absence of hardware costs simplifies setup and reduces upfront investment, while ongoing subscriptions ensure the system remains secure, available, and reliable for all users.

Design 3: Cloud-based Architecture with dedicated NAS Storage
Annual Cost

PROJECT INPUTS	Design 3	Cost
Hardware Implementation	Commercial NAS device Synology DS223 2 bay DiskStation® NAS	PHP 25,467.00
Domain name	Site TLD	HP 300.00
VPN service	protonVPN	PHP 2,160.00
Cloud Hosting Service	cloudways	PHP 7,920.00

TOTAL	PHP 35,847.00
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The total estimated cost for Design 3 of the INTRAK system is PHP 35,847.00. This design includes a commercial NAS device, specifically the Synology DS223 2-bay DiskStation® NAS, costing PHP 25,467.00, which provides secure, redundant, and centralized storage for internship documents. The domain name registration for a site TLD costs PHP 300.00, ensuring an accessible web address for students, instructors, and administrators. The VPN service uses ProtonVPN at PHP 2,160.00 to provide secure remote access to both cloud resources and the NAS. The cloud hosting service is provided by Cloudways at PHP 7,920.00, which hosts the web application and database, supporting reliable online access for all users.

Cloud hosting and VPN services contribute additional operational costs, while the domain name represents a minimal investment. This setup ensures that the INTRAK system maintains secure, centralized, and reliable access to both structured and unstructured internship data

Proposed Design	Total Annual Cost
Design 1	PHP 17,528.00
Design 2	PHP 18,040.00
Design 3	PHP 35,847.00

The estimated total annual costs for the proposed INTRAK system designs vary depending on the selected architecture. Design 1 has a total cost of PHP 17,528.00, covering the Raspberry Pi NAS-based setup, domain registration, VPN service, and Hostinger VPS hosting. Design 2, a fully cloud-based architecture without NAS integration, is slightly higher at PHP 18,040.00, reflecting the costs of a .com domain, ExpressVPN subscription, and Hostgator cloud hosting. Design 3, which incorporates a commercial NAS device (Synology DS223), a site TLD, ProtonVPN, and Cloudways hosting, has the highest total at PHP 35,847.00, mainly due to the investment in dedicated NAS hardware.

3.2.2 Tradeoff 2: Safety

3.2.3 Tradeoff 3: Cybersecurity Risk Assessment

The researchers applied Common Vulnerability Scoring System (CVSS) to provide a clear and standardized way in identifying vulnerabilities, assess possible threats, and measure risk levels. This approach was applied to the INTRAK System to ensure that the platform is secure, reliable, and capable of handling sensitive internship data.

Design 1: Cloud-Based with NAS Integration Using a Raspberry Pi

Design 1 presents moderate to high cybersecurity risk due to its reliance on a Raspberry Pi as a NAS device. From a CVSS perspective, vulnerabilities related to hardware limitations, patch management, and physical security increase the attack surface. Raspberry Pi devices are not enterprise-grade and may lack advanced security features such as hardware encryption acceleration, intrusion detection, and robust access control mechanisms.

Additionally, the NAS functionality hosted on a Raspberry Pi introduces risks related to availability and integrity, especially under heavy file access or concurrent users. If the device is misconfigured or not regularly updated, vulnerabilities may be exploitable remotely, increasing the CVSS exploitability score. While suitable for low-cost deployment, this design requires strict administrative controls and continuous monitoring to mitigate cybersecurity risks.

Second Design : Cloud-Based Angular–Spring Boot Web Platform for INTRAK

Design 2 demonstrates a lower cybersecurity risk compared to Design 1, primarily due to its fully cloud-based architecture and absence of on-premise or locally managed storage hardware. Using cloud hosting reduces risks related to physical access, hardware failure, and local misconfiguration. From a CVSS standpoint, this design benefits from improved availability and confidentiality, as cloud providers typically implement strong baseline security controls.

Third Design : Cloud-Based Architecture with Dedicated NAS Storage Using Python Full Stack

Design 3 offers the lowest cybersecurity risk among the three designs when evaluated using CVSS. The use of a dedicated commercial NAS device provides enhanced security features such as RAID redundancy, access control, encryption support, and centralized permission management. These features reduce the likelihood and impact of data loss, unauthorized access, and system downtime.

The researchers conclude that, from a CVSS perspective, Design 3 presents the lowest overall cybersecurity risk among the proposed solutions. Its controlled access to dedicated NAS storage, well-structured backend logic, and clear separation of system layers significantly reduce exploitability. Although risks related to misconfiguration and network exposure remain, their potential impact is minimized through enterprise-grade storage controls, access permissions, and redundancy mechanisms. Overall, this design achieves a balanced combination of security, reliability, and scalability, making it the most suitable option for managing sensitive internship records in an institutional environment.

3.2.4 Tradeoff 4: Environmental Tradeoff

The researchers evaluated the environmental impact of the three proposed designs of the INTRAK System, focusing on energy use, resource efficiency, and sustainability. Key aspects such as hardware, operational costs, storage management, and maintenance were analyzed to show how the system reduces energy consumption, limits electronic waste, and supports eco-friendly academic technology practices.

Design 1: Cloud-based with NAS Integration using a Raspberry Pi

ASPECT	DESCRIPTION	BENEFIT/COST	IMPACT ON ENVIRONMENT
Hardware	Raspberry Pi NAS	Lower Energy consumption / Compact design	Reduces electricity and carbon footprint
Operational Cost	Power usage for system operation	Reduced electricity bills	Less environmental impact from energy production
Storage and Document Management	Centralized NAS storage replaces paper records	Decrease need for printing and paper storage	Conserve natural resources and reduces paper waste
Maintenance and Upgrades	Lightweight hardware with modular components	Easier to maintain and replace	Minimize electronic waste compared to large servers
Sustainability	Long-term deployment with low power devices	Continues to provide secure storage with minimal energy	Support eco-friendly practices and sustainability goals

1. Hardware

The choice of a Raspberry Pi NAS as the primary hardware for the INTRAK System offers significant environmental advantages compared to conventional servers. The

Raspberry Pi consumes significantly less electricity due to its low-power design, which directly reduces the system's overall energy consumption. This not only lowers operational costs but also decreases the carbon footprint associated with powering the system, making it a more sustainable solution for the institution.

2. Operational Cost

Lower energy consumption directly translates into cost savings for the institution. Operating a Raspberry Pi NAS requires far less electricity than running standard desktop or server hardware. This reduction in energy usage not only benefits the institution financially but also indirectly contributes to environmental conservation by minimizing energy demand and the associated emissions from electricity generation.

3. Storage & Document Management

The centralized NAS storage system in INTRAK reduces the need for physical documentation. By storing internship records and student documents digitally, the system significantly decreases paper consumption. This helps conserve natural resources and reduces the waste generated by traditional paper-based systems, making the overall process more environmentally friendly.

4. Maintenance & Upgrades

The modular and lightweight design of the Raspberry Pi makes it easy to maintain and upgrade individual components without replacing the entire system. This reduces electronic waste, which is a common issue with traditional servers that require complete replacement when components fail or become outdated. Efficient maintenance practices extend the system's lifecycle while minimizing environmental impact.

5. Sustainability

The INTRAK System's design proposed by the researchers supports long-term environmental sustainability. The combination of low power consumption, digital document management, and modular hardware ensures that the system remains energy-efficient and eco-friendly over time. By adopting such technologies, the institution can achieve secure and reliable data management while also promoting greener IT practices and reducing its ecological footprint.

Design 2: Cloud-Based Angular–Spring Boot Web Platform for INTRAK

ASPECT	DESCRIPTION	BENEFIT/COST	IMPACT ON ENVIRONMENT
Hardware	N/A	No dedicated hardware needed.	Less e-waste, lower energy use.
Operational Cost	Cloud hosting, domain, VPN fees.	Predictable, scalable costs.	Moderate energy use, often from efficient data centers.
Storage and Document Management	Cloud storage with backup.	Reliable access, minimal local storage.	Reduces paper use; energy for data centers.
Maintenance and Upgrades	Software updates and monitoring handled in cloud.	Minimal IT effort.	Less on-site energy and resource use.
Sustainability	Uses virtualized, efficient cloud infrastructure.	Scalable, reliable system	Low local resource use; supports e-waste reduction.

Hardware: The system relies entirely on cloud servers, with no dedicated on-site hardware required. This eliminates the need for purchasing and maintaining physical servers, which reduces upfront costs. Users access the system through their own devices, such as desktops, laptops, or mobile devices. This setup also minimizes electronic waste and reduces the institution’s energy consumption.

Operational Cost:

The main recurring expenses include cloud hosting, domain registration, and VPN services. These costs are predictable and scale according to the number of users and storage requirements. While cloud servers consume energy, most providers operate energy-efficient and often green-certified data centers. This makes operational costs manageable while minimizing environmental impact.

Storage and Document Management:

All internship documents and reports are stored in cloud storage with backup and redundancy mechanisms. This ensures that files are always accessible and reduces the need for local storage infrastructure. Paper use is minimized since documents are fully digital, contributing to environmental conservation. The cloud storage also allows easy scaling as data volume grows without significant additional hardware.

Maintenance and Upgrades:

Software updates, security patches, and system monitoring are handled by the cloud provider. This reduces the workload for institutional IT staff and eliminates the need for on-site maintenance. Centralized updates also ensure that the system remains secure and reliable. By avoiding frequent on-site maintenance, energy and resource consumption are further reduced.

Sustainability:

The system leverages virtualized cloud infrastructure that is scalable and efficient. It reduces the need for physical resources and limits electronic waste compared to on-premise setups. Energy-efficient cloud operations help minimize the environmental footprint of system usage. Overall, this approach supports long-term sustainability while maintaining high reliability and performance.

Design 3: Cloud-based Architecture with dedicated NAS Storage using Python full stack

ASPECT	DESCRIPTION	BENEFIT/COST	IMPACT ON ENVIRONMENT
Hardware	Synology DS223 2-bay NAS along with cloud server	Secure, centralized storage; higher upfront cost	NAS uses electricity; reduces dependence on larger servers
Operational Cost	NAS, drives, cloud hosting, domain, VPN	Predictable recurring costs	Moderate energy use; some e-waste from NAS
Storage and Document Management	NAS stores documents with cloud backup	Reliable access, secure backups	Reduces paper use; NAS consumes moderate energy
Maintenance and Upgrades	Firmware updates, drive health checks	Prevents data loss; prolongs NAS life	Low resource use; reduces hardware replacement
Sustainability	Combines NAS with cloud backups	Scalable, durable, reliable	Lowers reliance on large data centers; limits e-waste

Hardware:

The system uses a Synology NAS to centralize storage, providing secure and organized data management. While the NAS requires an upfront investment and consumes electricity, it reduces reliance on large server infrastructures. Users can access files efficiently, ensuring smooth operations. Overall, the hardware setup balances cost, functionality, and reliability.

Operational Cost:

Costs include the NAS, storage drives, cloud hosting, domain, and VPN services. These are predictable and allow budgeting for both one-time and recurring expenses.

Although the NAS adds some energy consumption, the system avoids expensive on-premise server setups. The overall cost ensures a reliable, manageable infrastructure.

Storage & Document Management:

All internship documents are stored on the NAS with optional cloud backup for redundancy. This ensures safe and consistent access while minimizing data loss risks. Digital storage reduces the need for paper, making document management eco-friendly. The NAS also allows easy scaling as document volume grows.

Maintenance & Upgrades:

Routine firmware updates and drive checks help maintain NAS health and prolong its lifespan. Preventive maintenance reduces unexpected failures and ensures data integrity. While some IT oversight is needed, resource usage is minimal compared to replacing hardware frequently. This keeps the system reliable and cost-effective.

Sustainability:

The combination of NAS and cloud storage ensures long-term reliability and scalability. Energy-efficient cloud usage complements the NAS, reducing environmental impact compared to fully on-premise solutions. This setup limits e-waste while providing robust, sustainable operations for institutional use.

Environmental Analysis

The researchers tabulate the results of the proposed designs of the system. It shows that all approaches reduce paper usage and promote digital document management, thereby limiting waste. Design 1 (Raspberry Pi NAS-based) consumes very little power due to the low-energy Raspberry Pi, and its modular hardware allows upgrades without generating significant e-waste. Design 2 (Cloud-based Angular–Spring Boot) relies entirely on cloud infrastructure, minimizing on-site hardware and maintenance, though energy use depends on the data centers. Design 3 (Dedicated NAS with Python full stack) centralizes storage using a Synology NAS, which moderately increases local

energy use but provides long-term reliability, redundancy, and reduced paper consumption.

Among the three, Design 1 offers the most environmentally friendly solution due to its extremely low power consumption, minimal hardware footprint, and sustainable lifecycle management. Design 2 follows closely with low local resource use, while Design 3, although efficient and reliable, has higher energy demands because of the dedicated NAS device.

3.2.5 Tradeoff 5: Sustainability: Power Consumption, System Lifespan, and Maintainability

This researcher considers the sustainability trade-offs of the INTRAK System by examining power consumption, system lifespan, and maintainability of the three designs. These factors together highlight the system's environmental and operational sustainability.

Design 1: Cloud-Based with NAS Integration using a Raspberry Pi

Power Consumption: Design 1 has low power requirements due to the lightweight nature of the Raspberry Pi and USB-based storage. The system consumes significantly less electricity than traditional desktop servers or enterprise NAS devices, making it energy-efficient for continuous academic use. Low power consumption not only reduces operational costs but also lowers the environmental footprint, aligning with sustainability goals. Additionally, the reduced heat output from the Raspberry Pi helps maintain device reliability and eliminates the need for extensive cooling systems.

System Lifespan: The modularity of the Raspberry Pi and external storage components allows for easy upgrades or replacements without replacing the entire system. Individual parts such as SD cards or USB drives can be swapped before reaching the end of their expected life, extending the overall operational lifespan to

approximately 5–7 years. This design ensures long-term reliability while minimizing electronic waste, supporting sustainable IT practices within an institution.

Maintainability: Maintenance in this design is relatively simple. Software updates, system monitoring, and backup management can be performed with minimal technical effort. The lightweight setup also reduces the likelihood of system downtime, ensuring uninterrupted access to internship records. Routine checks of storage health and timely replacement of drives help maintain optimal system performance over time.

Design 2: Cloud-Based Angular–Spring Boot Web Platform for INTRAK

Power Consumption: In Design 2, local power usage is minimal because the system primarily relies on cloud infrastructure. The energy footprint depends largely on the efficiency of the cloud data centers, which typically employ advanced power management, virtualization, and optimized cooling solutions. Users' devices only handle lightweight tasks, further reducing local electricity consumption. This design is highly suitable for institutions that aim to lower energy use while providing scalable access to multiple users.

System Lifespan:

Because there is no dependency on on-site hardware, the system can operate continuously without the limitations of physical equipment aging. Cloud servers can be scaled, upgraded, or replaced by the service provider without any impact on the institution. This allows the system to remain up-to-date with the latest technologies and security measures, effectively giving it an indefinite lifespan as long as cloud services are maintained.

Maintainability:

Maintainability is one of the strongest advantages of this design. Cloud service providers handle software updates, backups, and server monitoring, reducing the need for dedicated IT staff. Any issues can often be resolved remotely without disrupting users. Centralized management simplifies administrative work, ensures consistent performance, and allows automated scaling to meet varying user demands.

Design 3: Cloud-Based Architecture with Dedicated NAS Storage using Python Full Stack

Power Consumption: Design 3 uses a dedicated Synology NAS, which requires moderate local power for continuous operation. While higher than a Raspberry Pi or fully cloud-based system, this power consumption is justified by the reliable storage, redundancy, and high availability it provides. The combination of local NAS and optional cloud backup balances performance with energy use, ensuring critical data is always accessible while maintaining moderate operational costs.

System Lifespan: The Synology NAS is designed for long-term use, with features such as RAID configurations, regular firmware updates, and health monitoring of drives. With proper care, the system can remain operational for many years, and failing components can be replaced individually without replacing the entire NAS. This extends the lifespan of the system significantly compared to traditional all-in-one hardware setups and ensures data remains protected over the long term.

Maintainability: This design requires more active maintenance than a fully cloud-based solution. IT personnel must monitor NAS health, perform periodic backups, update software, and replace drives as necessary. However, the modularity of the NAS allows maintenance tasks to be performed without major disruptions to the system. While more resource-intensive, this approach ensures data integrity, resilience against hardware failure, and reliable access for all users. Maintainability in an academic setting.

3.3 Influence of the Design Tradeoffs on the Final Design

Economic Trade-off: The researchers prioritized cost-effectiveness by selecting a Raspberry Pi NAS-based storage solution instead of a traditional server. This choice reduced initial investment and operational expenses while still providing sufficient performance for managing student internship data.

Safety Trade-off: Security measures, such as user authentication and access controls, were implemented to protect sensitive information. The design ensured that these safety features did not complicate system use, maintaining a balance between protection and user convenience.

Risk Trade-off: The system focused on mitigating high-impact vulnerabilities, such as unsecured NAS directories or outdated software packages, while lower-risk issues received fewer resources. This allowed efficient allocation of security measures without overcomplicating the system.

Environmental Trade-off: The INTRAK System was designed with low-power hardware and digital document management to minimize energy consumption and reduce paper usage. These measures limit the system's carbon footprint and contribute to sustainable IT practices.

Sustainability Trade-off: Long-term maintainability and system lifespan were emphasized through modular software design, well-documented code, and routine maintenance. These practices ensure the system remains functional, adaptable, and reliable over time, supporting institutional needs without frequent overhauls.

By carefully considering these trade-offs, the researchers developed a final INTRAK design that is cost-efficient, secure, environmentally responsible, and sustainable, providing a reliable platform for managing internship data in an academic setting.

Summary of Normalized Values and Design Selection

The trade-off scores are summarized in the following table:

Design Criteria	Weight	Design 1 Cloud + RPI NAS	Design 2 Full Cloud	Design 3 Cloud + Commercial NAS
Economic Cost	20%	8	7	10
Safety/Security	25%	9	6	8
Risk/Reliability	20%	9	8	6
Environmental	15%	7	9	8

Sustainability	20%	9	8	6
Weighted Score	100%	8.50	7.45	7.60

Weighting Rationale:

Safety/Security (25%): Given the highest weight because student records must be kept private and protected, in line with the Data Privacy Act.

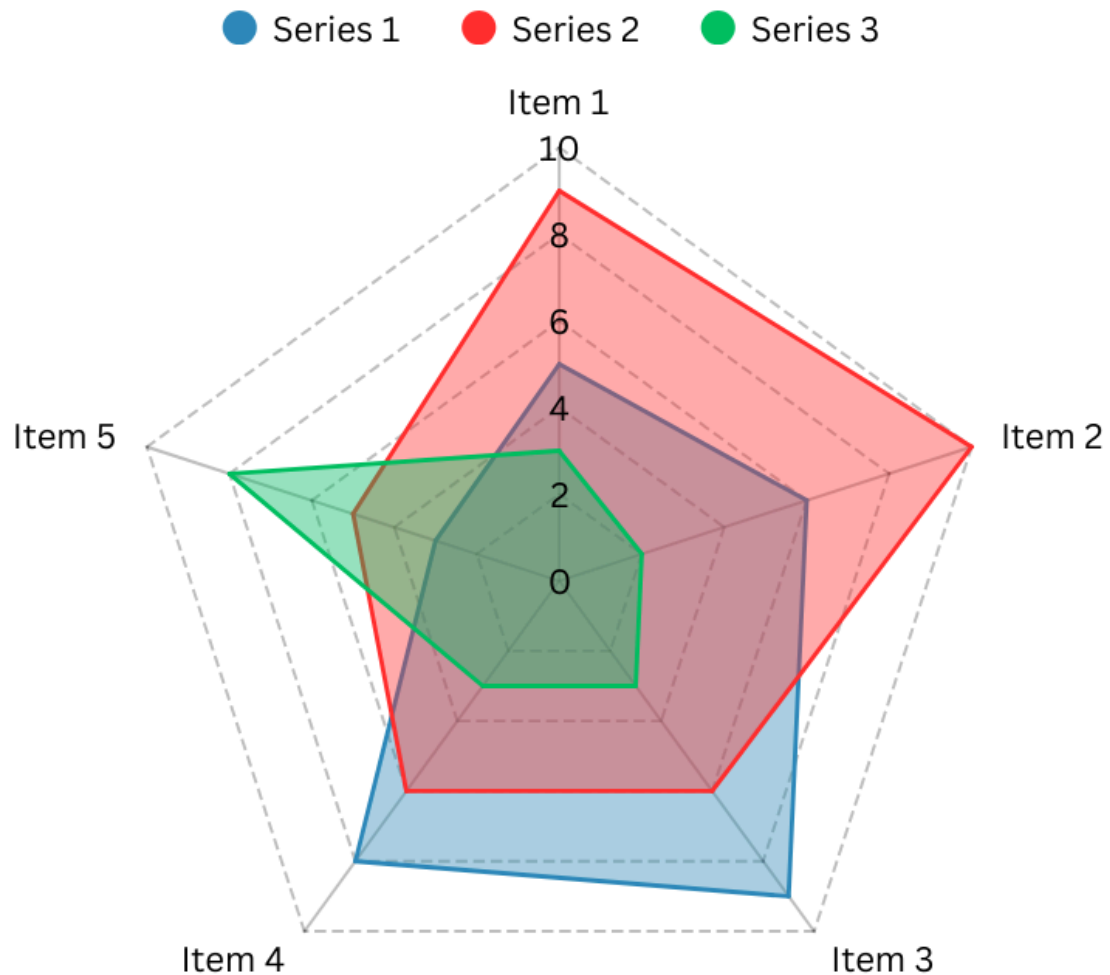
Economic (20%): Important due to budget limits, but cost is not the main factor in system design.

Risk/Reliability (20%): High weight because the system needs to be available and reliable for daily academic use.

Sustainability (20%): Important to ensure the system can operate and be maintained over many years.

Environmental (15%): Considered important, but less critical than security, reliability, and system performance.

RADAR CHART



Design 1: Cloud-based with NAS Integration using a Raspberry Pi

Design 1 demonstrates strong and consistent performance across the most critical criteria. It scores highest in Safety/Security (9), Risk/Reliability (9), and Sustainability (9), reflecting its suitability for protecting sensitive student data and supporting long-term operations. While its scores for Economic Cost (8) and Environmental impact (7) are slightly lower, the design remains well-balanced overall. On the radar chart, this creates a large, evenly spread polygon, visually showing its robustness and alignment

with institutional priorities. The weighted score of 8.50 confirms that Design 1 is the top choice for the project.

Design 2: Cloud-Based Angular–Spring Boot Web Platform for INTRAK

Design 2 performs well in Environmental (9) and Economic Cost (7), indicating lower hardware requirements and simplified operational management. However, it scores lower in Safety/Security (6) and slightly lower in Sustainability (8), reflecting the risks of relying on external cloud providers and limited local control of sensitive data. On the radar chart, this creates an irregular shape with noticeable dips in security-related areas, signaling vulnerabilities. Its weighted score of 7.45 shows that, while cost-effective and environmentally friendly, Design 2 is less suitable for institutional priorities where data privacy and reliability are critical.

Design 3: Cloud-based Architecture with dedicated NAS Storage using Python full stack

Design 3 shows high scores in Economic Cost (10) and Environmental impact (8), highlighting efficiency and lower operational impact. However, it scores lower in Risk/Reliability (6) and Sustainability (6), suggesting potential weaknesses in long-term system stability and reliability. The radar chart shows a narrower, uneven polygon, with peaks in cost and environment but significant gaps in critical operational areas. With a weighted score of 7.60, Design 3 performs slightly better than Design 2 overall, but it is less balanced and less robust than Design 1 in addressing key institutional needs.

Conclusion

The radar chart shows that Design 1 (Cloud + RPI NAS) is the most balanced and robust option, scoring highest in security, reliability, and sustainability, making it well-suited for institutional priorities. Design 2 (Full Cloud) is strong in environmental impact and cost but weaker in security and long-term control, making it less suitable. Design 3 (Cloud + Commercial NAS) performs well in cost and environmental criteria but has

lower reliability and sustainability scores. The chart confirms that Design 1 is the optimal choice for the project.

Design Selection Decision

Design 1 (Hybrid Cloud with NAS Storage) is chosen as the best solution, with a weighted score of 8.50 out of 10, performing better than the other designs.

This decision is based on the following reasons:

1. **Meets Project Constraints:** Design 1 keeps student data under institutional control while still allowing secure remote access when needed.
2. **Strong Security:** The design provides high data protection, which is essential for safeguarding student records and complying with data privacy requirements.
3. **Balanced Performance:** Design 1 performs well across all evaluation criteria and does not show major weaknesses. It avoids the limited scalability of Design 3 and the dependence on external providers seen in Design 2.
4. **Good Institutional Fit:** The design matches common university IT practices and works well with existing network and backup systems.
5. **Supports Future Growth:** The NAS-based setup allows easy expansion as more users, departments, and data are added in the future.

Although Design 3 scores highest in terms of cost efficiency, it performs poorly in scalability-related areas such as risk and sustainability. This makes it unsuitable for a system expected to grow over time.

Design 2, while offering advantages like automatic scaling and reduced maintenance, does not meet the requirement for local data storage and creates reliance on external service providers, which is not acceptable for an academic institution.

Influence of the Design Trade-Off on the Final Design

The trade-off analysis had a direct impact on how the final INTRAK system was designed and implemented.

Storage Architecture Decision

The choice to use NAS storage strongly influenced the overall system structure. Uploaded files are stored on the NAS using the SMB protocol, while the web system uses secure HTTPS connections for remote access. The system includes the needed network permissions and file access tools to communicate with the NAS. Error handling is designed to manage storage-related issues such as network interruptions or lack of available storage space.

API Design

The storage approach affected how the system's API was designed. During file uploads, the backend first saves the file to the NAS and then stores the file details in the PostgreSQL database. This two-step process ensures that both the file and its information are properly recorded. Transaction controls are used to keep the file storage and database records consistent.

Security Implementation

Security considerations played a major role in the final design. All file actions require proper user authorization before they are allowed. The system records file access activities, including the user, time, and type of action performed. Access to the NAS is limited to system service accounts, preventing direct access by students or faculty. Secure connections, firewalls, and VPN requirements are enforced for all remote access.

Capacity Planning

Sustainability considerations guided storage capacity planning. The NAS was configured with an initial usable capacity of 12TB to support projected system growth over five years. This estimate is based on expected document sizes and user numbers, with additional space included to allow safe long-term operation.

Infrastructure Requirements

The trade-off analysis also influenced hardware selection. The final design uses enterprise-grade NAS hardware with RAID 5 for improved reliability. Consumer-grade devices were excluded because they do not meet the system's reliability and risk requirements.

Backup Strategy

Using NAS storage allows easy integration with existing institutional backup systems. The backup plan includes daily incremental backups and weekly full backups of stored documents. This approach supports long-term data protection and recovery needs.

Deployment Architecture

The NAS is placed on the same network as the application server to improve file transfer speed and reduce delays. Gigabit Ethernet connectivity ensures smooth system performance. A secure internet gateway is used to allow controlled remote access for authorized external users.

Rejected Alternative Features

Pure cloud-based storage was not selected, and features such as public content delivery networks were excluded. Instead, remote access needs are met through the hybrid NAS setup, allowing secure access while keeping institutional data under local control.

3.4 Sensitivity Analysis

Sensitivity analysis evaluates the robustness of the design selection decision under varying weightings of the evaluation criteria. By systematically adjusting criterion weights, the analysis determines whether Design 1 remains optimal across different priority scenarios or if the decision is fragile to weight changes.

Ten sensitivity iterations explore different weighting schemes, representing alternative stakeholder priority rankings. Each iteration maintains total weight of 100% while varying individual criterion emphasis.

Iteration	Weighting	Design 1	Design 2	Design 3
1	Economic-Focused (10-9-8-7-6)	8.60	7.05	8.50
2	Security-Focused (6-10-7-8-9)	9.05	7.60	7.25
3	Environmental-Focused (6-8-10-7-9)	8.45	8.20	6.95
4	Reliability-Focused (9-7-10-8-6)	8.80	7.75	7.35
5	Sustainability-Focused (8-7-6-9-10)	8.70	7.90	7.05
6	Balanced Uniform (10-9-6-7-8)	8.60	7.60	7.60
7	Cost-Minimization (7-9-8-6-10)	8.35	7.00	8.85
8	Risk-Averse (8-6-9-10-7)	8.75	7.65	7.30
9	Cloud-Friendly (9-10-8-6-7)	8.10	7.90	7.50
10	Operations-Focused (7-10-9-6-8)	8.90	7.55	7.10

Design 1: Cloud-based with NAS Integration using a Raspberry Pi

Design 1 consistently performs well across most weighting scenarios. It scores highest in security-focused iterations (9.05 in Iteration 2) and remains strong in reliability, sustainability, and balanced scenarios. Even in cost-focused iterations, its score is competitive (8.60 in Iteration 1), showing that it maintains good performance across different priorities. The radar chart would show a large, fairly even polygon, indicating

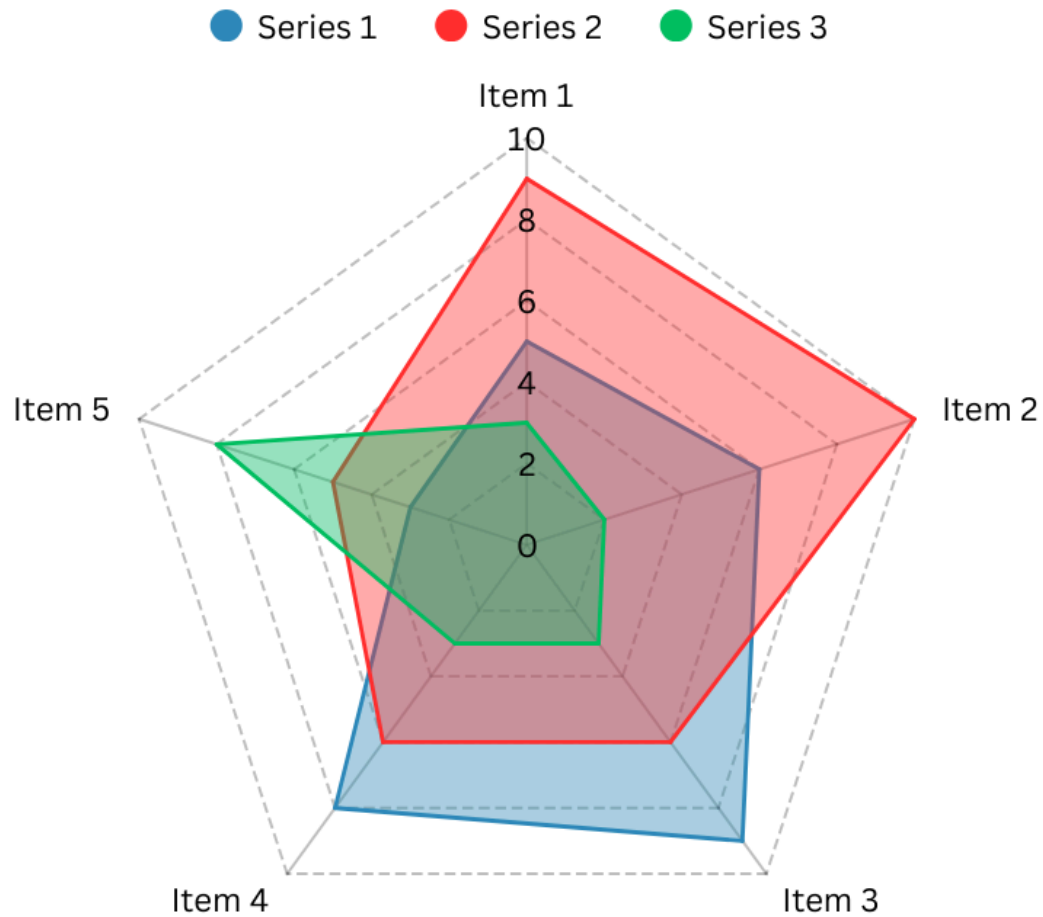
a robust and stable design that aligns with institutional priorities and regulatory requirements.

Design 2: Cloud-Based Angular–Spring Boot Web Platform for INTRAK

Design 2 performs moderately across all iterations. It scores relatively higher in environmental-focused (8.20 in Iteration 3) and balanced iterations, but it consistently underperforms in economic-focused (7.05 in Iteration 1) and security-focused scenarios (7.60 in Iteration 2). On the radar chart, the polygon would appear smaller and irregular, reflecting its vulnerability in critical areas like security and risk management, making it less suitable for an institution where data protection is a priority.

Design 3: Cloud-based Architecture with dedicated NAS Storage using Python full stack

Design 3 shows strong performance in cost-sensitive iterations (8.85 in Iteration 7) but lower scores in security, sustainability, and operations-focused scenarios (as low as 7.05 in Iteration 5 and 7.10 in Iteration 10). The radar chart would reveal a spiky, uneven shape, highlighting high variation depending on criteria weighting. This indicates that while Design 3 may be attractive for cost efficiency, it is less balanced and less robust compared to Design 1.



Conclusion

The radar chart confirms that Design 1 is the most balanced and reliable design, maintaining high scores across most priority scenarios. Its polygon is broad and even, representing consistent performance and robustness to changes in weighting. Design 2 is weaker and irregular, particularly in security and risk, while Design 3 is variable, performing well in cost-focused scenarios but poorly in reliability and sustainability. The radar chart analysis supports the selection of Design 1 (Cloud + RPI NAS) as the optimal solution for institutional needs and project requirements.

Minimum Margin of Victory:

Design 1's smallest advantage appears in Iteration 1, where the focus is mainly on cost (8.60 vs. 8.50), a difference of only 0.10 points. However, this case places too much

importance on cost and reduces the importance of security and sustainability, which does not reflect the institution's actual priorities.

Maximum Margin of Victory:

Design 1's largest advantage occurs in Iteration 2, which focuses on security (9.05 vs. 7.60), resulting in a difference of 1.45 points. This scenario closely matches the institution's needs, where protecting data is the top concern.

Stability Analysis:

Design 1 shows more consistent results across all test cases. Its score changes only slightly compared to Design 2 and Design 3. Design 3 shows large changes in score, meaning its performance depends too much on cost-related weighting.

Constraint Violation Impact

Design 2 does not meet the requirement for local data storage, which is a strict project rule. Because of this, it cannot be selected, even if it were to achieve a higher numerical score.